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**ADDITIONAL MATHEMATICS** **4037/11**

Paper 1 **May/June 2010**

**2 hours**

Additional Materials: Answer Booklet/Paper

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***Mathematical Formulae***

## 1. ALGEBRA

*Quadratic Equation*

For the equation $ax^2 + bx + c = 0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

*Binomial Theorem*

$$(a + b)^n = a^n + \binom{n}{1} a^{n-1} b + \binom{n}{2} a^{n-2} b^2 + \ldots + \binom{n}{r} a^{n-r} b^r + \ldots + b^n,$$

where $n$ is a positive integer and $\binom{n}{r} = \frac{n!}{(n-r)!r!}$.

## 2. TRIGONOMETRY

*Identities*

$$\sin^2 A + \cos^2 A = 1.$$

$$\sec^2 A = 1 + \tan^2 A.$$

$$\operatorname{cosec}^2 A = 1 + \cot^2 A.$$

*Formulae for $\Delta ABC$*

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.$$

$$a^2 = b^2 + c^2 - 2bc \cos A.$$

$$\Delta = \frac{1}{2} bc \sin A.$$

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**1** Differentiate with respect to $x$

**(i)** $\sqrt{1 + x^3}$,

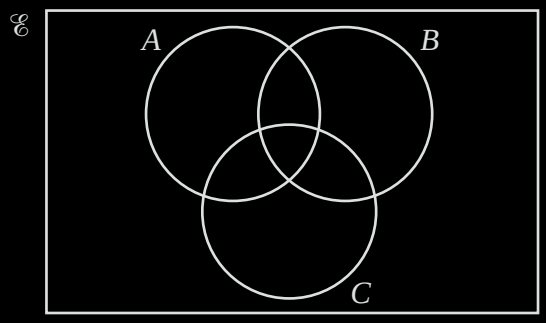
**(ii)** $x^2 \cos 2x$. [4]

**2** **(i)** Find the first 3 terms of the expansion, in ascending powers of $x$, of $(1 + 3x)^6$. [2]

**(ii)** Hence find the coefficient of $x^2$ in the expansion of $(1 + 3x)^6 (1 – 3x – 5x^2)$. [3]

**3** Find the set of values of $k$ for which the equation $x^2 + (k – 2)x + (2k – 4) = 0$ has real roots. [5]

**4** **(a)**




**(i)** Copy the Venn diagram above and shade the region that represents $(A \cap B) \cup C$. [1]

**(ii)** Copy the Venn diagram above and shade the region that represents $A' \cap B'$. [1]

**(iii)** Copy the Venn diagram above and shade the region that represents $(A \cup B) \cap C$. [1]

**(b)** It is given that the universal set $\mathscr{E} = \{x : 2 \leqslant x \leqslant 20, x \text{ is an integer}\}$,
$X = \{x : 4 < x < 15, x \text{ is an integer}\}$,
$Y = \{x : x \geqslant 9, x \text{ is an integer}\}$,
$Z = \{x : x \text{ is a multiple of } 5\}$.

**(i)** List the elements of $X \cap Y$. [1]

**(ii)** List the elements of $X \cup Y$. [1]

**(iii)** Find $(X \cup Y)' \cap Z$. [1]

**5** Solve the equation $3x(x^2 + 6) = 8 – 17x^2$. [6]

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**6** Given that $\log_8 p = x$ and $\log_8 q = y$, express in terms of $x$ and/or $y$

**(i)** $\log_8 \sqrt{p} + \log_8 q^2$,

**(ii)** $\log_8 \left(\frac{q}{8}\right)$, [2]

**(iii)** $\log_2 (64p)$. [3]

**7** The function f is defined by

$$f(x) = (2x + 1)^2 - 3 \quad \text{for } x \geqslant -\frac{1}{2}.$$

Find

**(i)** the range of f, [1]

**(ii)** an expression for $f^{-1}(x)$. [3]

The function g is defined by

$$g(x) = \frac{3}{1 + x} \quad \text{for } x > -1.$$

**(iii)** Find the value of $x$ for which $fg(x) = 13$. [4]

**8** **(a)** Solve the equation $(2^{3-4x})(4^{x+4}) = 2$. [3]

**(b)** **(i)** Simplify $\sqrt{108} - \frac{12}{\sqrt{3}}$, giving your answer in the form $k\sqrt{3}$, where $k$ is an integer. [2]

**(ii)** Simplify $\frac{\sqrt{5}+3}{\sqrt{5}-2}$, giving your answer in the form $a\sqrt{5} + b$, where $a$ and $b$ are integers. [3]

**9** **(a)** Variables $x$ and $y$ are related by the equation $y = 5x + 2 - 4e^{-x}$.

**(i)** Find $\frac{dy}{dx}$. [2]

**(ii)** Hence find the approximate change in $y$ when $x$ increases from 0 to $p$, where $p$ is small. [2]

**(b)** A square of area $A\,\text{cm}^2$ has a side of length $x\,\text{cm}$. Given that the area is increasing at a constant rate of $0.5\,\text{cm}^2\,\text{s}^{-1}$, find the rate of increase of $x$ when $A = 9$. [4]

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**10** Solve

**(i)** $4\sin x = \cos x$ for $0° < x < 360°$,

**(ii)** $3 + \sin y = 3\cos^2 y$ for $0° < y < 360°$, [5]

**(iii)** $\sec\left(\frac{z}{3}\right) = 4$ for $0 < z < 5$ radians. [3]

**11** Answer only **one** of the following two alternatives.

**EITHER**

A curve has equation $y = \frac{\ln x}{x^2}$, where $x > 0$.

**(i)** Find the exact coordinates of the stationary point of the curve. [6]

**(ii)** Show that $\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$ can be written in the form $\frac{a\ln x + b}{x^4}$, where $a$ and $b$ are integers. [3]

**(iii)** Hence, or otherwise, determine the nature of the stationary point of the curve. [2]

**OR**

A curve is such that $\frac{\mathrm{d}y}{\mathrm{d}x} = 6\cos\left(2x + \frac{\pi}{2}\right)$ for $-\frac{\pi}{4} \leqslant x \leqslant \frac{5\pi}{4}$. The curve passes through the point $\left(\frac{\pi}{4}, 5\right)$.

Find

**(i)** the equation of the curve, [4]

**(ii)** the $x$-coordinates of the stationary points of the curve, [3]

**(iii)** the equation of the normal to the curve at the point on the curve where $x = \frac{3\pi}{4}$. [4]

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